Entergy

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1CAN100703

October 22, 2007

U.S. Nuclear Regulatory Commission
Attn: Document Control Desk
Washington, DC 20555

SUBJECT: License Amendment Request
Technical Specification Changes and Analyses Relating to Use of Alternate Source Term
Arkansas Nuclear One, Unit 1
Docket No. 50-313
License No. DPR-51

References:
1. Entergy letter to NRC dated October 22, 2007, Technical Specification Changes Control Room Envelope Habitability in Accordance With TSTF-448, Revision 3, Using the Consolidated Line Item Improvement Process (1CAN100704)
2. Entergy letter to NRC dated February 14, 2007, Supplemental Response to GL 2003-01 Regarding Control Room Habitability (0CAN020701)

Dear Sir or Madam:

10 CFR 50.67(b) states: *A licensee who seeks to revise its current accident source term in design basis radiological consequence analyses shall apply for a license amendment under Sec. 50.90. The application shall contain an evaluation of the consequences of applicable design basis accidents previously analyzed in the safety analysis report.*

Therefore, in accordance with the provisions of 10 CFR 50.90, Entergy Operations, Inc. (Entergy) is submitting a request for license amendment to implement an alternate source term (AST) for Arkansas Nuclear One, Unit 1 (ANO-1). In support of AST implementation and also in accordance with the provisions of 10 CFR 50.90, Entergy is including a request for an amendment to the Technical Specifications (TS) for ANO-1. The proposed amendment will modify TS requirements related to the use of an AST associated with accident offsite and control room dose consequences. Implementation of AST supports adoption of the control room envelope habitability controls in accordance with Technical Specification Task Force (TSTF)-448, Revision 3 (Reference 1). Entergy committed to submit a proposal to develop an AST, adopt TSTF-448, and retire current compensatory measures relating to control room habitability in letter dated February 14, 2007 (Reference 2).

A001
A102
NRR

A markup of affected Technical Specification (TS) pages is included in Attachment 2. Attachment 3 includes summary tables of dose analyses input using AST methodology of NRC Regulatory Guide (RG) 1.183. Attachment 4 includes a summary of the results of dose analyses for the events that are expected to produce the most limiting dose consequences, in accordance with RG 1.183.

The proposed change has been evaluated in accordance with 10 CFR 50.91(a)(1) using criteria in 10 CFR 50.92(c) and it has been determined that the change involves no significant hazards consideration. The bases for these determinations are included in Attachment 1.

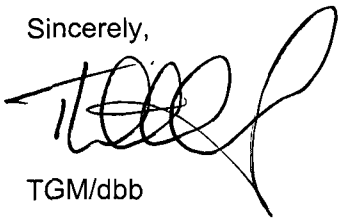
Entergy requests acceptance of the ANO-1 use of AST as discussed in Attachment 1 and approval of the proposed TS changes by July 1, 2008, concurrent with NRC acceptance/approval of the proposed ANO-1 TSTF-448, Revision 3, application (Reference 1). Approval of both applications will retire current compensatory measures relating to control room habitability for ANO-1. Once approved, the amendment shall be implemented within 60 days. Although this request is neither exigent nor emergency, your prompt review is requested.

There are no new commitments included in this letter.

If you have any questions or require additional information, please contact David Bice at 479-858-5338.

I declare under penalty of perjury that the foregoing is true and correct. Executed on October 22, 2007.

Sincerely,



TGM/dbb

Attachments:
1. Analysis of Proposed Change
2. Proposed Technical Specification Changes (mark-up)
3. Summary Tables of Dose Analyses Input
4. Summary Results of Dose Analyses

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Attachment 1

1CAN100703

Analysis of Proposed Change

1.0 DESCRIPTION

This letter is a request to amend Operating License DPR-51 for Arkansas Nuclear One, Unit 1 (ANO-1). The proposed change will implement an alternative source term (AST) for determining accident offsite and control room doses. The AST is being adopted principally as part of the Entergy Operations, Inc. (Entergy) response to Generic Letter 2003-01, Control Room Habitability (References 1 and 8). A license amendment is required for AST implementation in accordance with 10 CFR 50.67(b) which states:

A licensee who seeks to revise its current accident source term in design basis radiological consequence analyses shall apply for a license amendment under Sec. 50.90. The application shall contain an evaluation of the consequences of applicable design basis accidents previously analyzed in the safety analysis report.

The proposed amendment will also modify Technical Specification (TS) requirements related to the use of an AST associated with accident offsite and control room dose consequences. Implementation of AST supports adoption of the control room envelope habitability controls in accordance with Technical Specification Task Force (TSTF)-448, Revision 3.

A markup of affected TS pages is included in Attachment 2. Attachment 3 includes summary tables of dose analyses input using AST methodology of NRC Regulatory Guide (RG) 1.183. Attachment 4 includes a summary of the results of dose analyses for the following events that are expected to produce the most limiting dose consequences:

- Loss of Coolant Accident (LOCA)
- Control Rod Ejection Accident (CREA)
- Steam Generator Tube Rupture (SGTR)
- Outside Containment Main Steam Line Break (MSLB)
- Fuel Handling Accident (FHA)

2.0 PROPOSED CHANGE

The change to the ANO-1 licensing basis involves the adoption of an AST for calculating accident doses to control room personnel and offsite receptors. The following TS revisions are requested to support assumptions associated with the new AST analyses:

TS 3.4.12 - The allowable equilibrium specific activity of the RCS is reduced to 1.0 µCi/gm dose equivalent I-131. In addition, Action B is revised to limit any iodine spike to 60 µCi/gm dose equivalent I-131. The revised action is consistent with the Standard Technical Specifications (STS) of NUREG-1430, Revision 3.1 considering NRC approval of Technical Specification Task Force (TSTF) 490, Revision 0, published in an NRC Notice of Availability dated March 15, 2007.

TS 3.7.4 - The allowable equilibrium specific activity of the secondary coolant is reduced to 0.1 µCi/gm dose equivalent I-131.

3.0 BACKGROUND

The control rooms for ANO-1 and Arkansas Nuclear One, Unit 2 (ANO-2) adjoin each other. The walls between the control rooms contain louvers resulting in a common control room envelope (CRE). The normal air conditioning systems for the two control rooms are separate. The safety-related emergency ventilation systems, provided for control room habitability (CRH), are shared by the common CRE. The CRE is designed and maintained to be as airtight, as practicable. The two units have separate licensing basis for acceptable inleakage based on differing accident analyses.

Two normal ventilation systems, one for ANO-1 and one for the ANO-2, are provided to manage the control room environment during routine operations. The air intake into each system is continuously monitored for radiation, chlorine, and smoke. Upon receiving a high radiation or high chlorine concentration signal from the normal air intakes, the CRE is isolated, except for filtered outside air used for pressurization to minimize unfiltered air inleakage. The arrangement assures redundancy of the monitoring system.

The safety-related control room emergency ventilation system (CREVS) provided for the CRH is comprised of two trains. Each train includes a CREVS and a control room emergency air conditioning system (CREACS). Each train of the CREVS has a filter unit that is located outside the ANO-1 section of the common control room. Each filter unit includes a centrifugal fan, a roughing filter, a high efficiency particulate (HEPA) filter, and a charcoal absorber. Besides recirculation and filtration of control room air, filtered outside makeup air is also provided to pressurize the control room in order to minimize unfiltered air inleakage into the control room following control room isolation. The CREVS was originally designed to reduce the potential control room operator dose from a radiological accident to within General Design Criterion (GDC) 19 limits, based only upon an unfiltered inleakage rate of 10 standard cubic feet per minute (scfm) due to ingress and egress from the control room during an event. However, the tracer gas testing performed in November 2001 indicates the actual unfiltered inleakage rate (30 scfm, not including ingress and egress) is in excess of this value.

As stated above, the original analysis of control room habitability for ANO-1 was established assuming 10 scfm unfiltered inleakage into the control room with a closed recirculation system that continuously recirculated air inside the control room through charcoal filters. During the license amendment review for power uprate on ANO-2, Entergy performed a tracer gas test on the ANO control room envelope to determine the actual unfiltered inleakage into the control room.

A calculation was performed to determine maximum allowable inleakage without compensatory measures, as well as the maximum allowable inleakage with possible compensatory measures credited. Consideration was given to administratively lowering the ANO-1 allowable reactor building leakage, issuance of potassium iodide (KI) to control room operators, and use of self-contained breathing apparatus (SCBA). This calculation reviewed accidents for which control room doses had been calculated to demonstrate compliance with GDC 19 in the ANO licensing basis. For ANO-1, consideration was given to the following accidents for which control room doses had been assessed:

- The maximum hypothetical design basis large break LOCA (or MHA)
- The FHA

The control room dose consequences of the other accidents described in Chapter 14 of the ANO-1 Safety Analysis Report (SAR) had not been previously calculated. The design/licensing basis for the ANO-1 MSLB, the SGTR, and the Loss of Load accidents does not assume iodine spiking or concurrent loss of offsite power. The release terms and durations of these accidents have historically been judged such that the control room doses from these accidents are bounded by those of the MHA. For the CREA (as described in the ANO-1 SAR), the offsite doses at the Exclusion Area Boundary (EAB) and Low Population Zone (LPZ) were a fraction (< 10%) of the doses resulting from the MHA. Therefore, the control room doses from a CREA were judged to be bounded by those of the MHA at the same assumed unfiltered inleakage rate.

The results of the analysis demonstrated that GDC 19 acceptance criteria could be met for unfiltered inleakage of ≤ 26 scfm. With the issuance of KI as a compensatory measure, Entergy calculations demonstrated that GDC 19 limits would be met for unfiltered inleakage of up to 340 scfm for the ANO-1 MHA. Therefore, ANO-1 is presently in an operable but degraded condition, with a measured inleakage of 40 scfm (including 10 scfm due to ingress/egress), which is in excess of the 10 scfm described in the SAR and the 26 scfm determined in the operability evaluation to be the maximum acceptable unfiltered inleakage to meet GDC 19 limits without compensatory measures.

Based on the results of the tracer gas testing, and in order to retire the current compensatory measure, Entergy has decided to implement an AST for ANO-1 in the calculation of accident doses to control room personnel. The revised dose consequence analyses have been performed for the event scenarios described in Regulatory Guide 1.183. The AST will also be adopted for calculating offsite accident dose consequences.

5.0 TECHNICAL ANALYSIS

The AST and methodology described in NUREG-1465, "Accident Source Terms for Light-Water Nuclear Power Plants" (Reference 2) and in RG 1.183, "Alternative Source Terms for Evaluating Design Basis Accidents at Nuclear Power Reactors" (Reference 3), provide regulatory guidance for the implementation of the AST. Revision of a plant licensing basis from the TID-14844 source term to an alternative source term involves the preparation of dose consequence analyses. Demonstration that the results satisfy the regulatory acceptance criteria and NRC approval of the requested change establishes the acceptability of the use of the AST.

Entergy has prepared dose consequence analyses of the eight event scenarios identified below. A summary of the major inputs to each of these analyses is provided in Attachment 3. The analyses included evaluation of the worst-case 2-hour EAB offsite doses, the duration LPZ offsite doses, and control room doses. The results for all of the analyzed events meet the regulatory acceptance criteria of 10 CFR 50.67 and RG 1.183.

Assuming a minimum control room (CR) unfiltered inleakage of 82 scfm, the dose results (Total Effective Does Equivalent or TEDE, in rem) of the calculations performed are as follows:

Accident	EAB	LPZ	CR
LOCA	10.49	2.56	3.77
CREA (containment leakage)	4.73	2.28	3.40
CREA (primary-secondary leakage)	3.03	1.64	4.95
MSLB (pre-accident iodine spike)	0.45	0.19	1.84
MSLB (concurrent iodine spike)	2.07	1.05	3.72
SGTR (pre-accident iodine spike)	2.20	0.37	2.33
SGTR (concurrent iodine spike)	1.26	0.23	1.00
FHA	1.40	0.25	1.00

The results show that the CREA, via the primary-secondary leakage pathway, becomes bounding for the ANO-1 control room (least margin to acceptance criteria). The MHA thyroid consequences were previously bounding for the control room as currently reported in the ANO-1 SAR. The FHA results are without Emergency Safeguards Features (ESF) or purge filtration credit.

As stated above, the new dose consequence analyses have been performed for the various accidents identified above using RG 1.183. No exceptions to the guidance contained in RG 1.183 have been taken. In addition, to the extent practical, the current ANO-1 licensing basis analyses input and assumptions have been utilized, provided no conflict was created with the guidance of RG 1.183. Major exceptions to the use of the current ANO-1 licensing basis are provided below:

Spray Fraction

The actual net free volume of the ANO-1 reactor building and its sprayed fraction were updated. Although the results of these calculations had an insignificant impact on doses, the previous sprayed-fraction calculation utilized information that was developed while the plant was still under construction. The revised calculation provides a better basis for the values used in the analyses. The revised sprayed fraction is 89%; unsprayed is 11%.

Atmospheric Dispersion Factors

The control room χ/Q values from the calculations previously used for the ANO-2 extended power uprate analyses (already reviewed and approved by the NRC for use on that unit) were used in the new ANO-1 calculations. Attachment 3 provides the updated χ/Q values used in the new ANO-1 analyses.

RCS and Secondary Specific Activity

The assumed equilibrium Reactor Coolant System (RCS) specific activity was reduced to 1.0 µCi/gm dose equivalent I-131. Consideration was also given to a pre-existing iodine spike of 60 µCi/gm dose equivalent I-131. The assumed equilibrium secondary specific activity was reduced to 0.1 µCi/gm dose equivalent I-131. Proposed changes to associated TSs 3.4.12 and 3.7.4 are subsequently included in this license amendment request.

RCS Leakage Rate

The maximum primary-to-secondary leak rate allowed by the ANO-1 TSs was used in the CREA and SGTR analyses. This value is 150 gallons per day (gpd) per Steam Generator (SG). The MSLB analyses continue to use a total leak rate of 1 gpm.

Fuel Handling Delay Time

The delay time after shutdown prior to handling fuel was decreased from 100 hours to 72 hours for the FHA analyses.

Based on the above results, the GDC 19 acceptance criteria are met for all ANO-1 accidents with a control room unfiltered inleakage of 82 scfm. Since this value exceeds the actual measured unfiltered inleakage of 30 scfm by greater than 10 scfm (to account for ingress/egress), the control room will no longer require any compensatory measure to ensure design and licensing basis compliance. Therefore, reliance on issuance of KI and credit for this compensatory measure will be retired upon NRC acceptance of this license amendment request.

5.0 REGULATORY ANALYSIS

5.1 Applicable Regulatory Requirements/Criteria

The proposed changes have been evaluated to determine whether applicable regulations and requirements continue to be met.

Entergy has determined that the proposed changes do not require any exemptions or relief from regulatory requirements, other than the existing Technical Specifications (TSs) with regard to Reactor Coolant System (RCS) and Secondary System specific activities. As discussed in Section 2.0 above, Entergy proposes to revise TS 3.4.12, RCS Specific Activity, and TS 3.7.4, Secondary Specific Activity, to be consistent with the activities assumed in the Steam Generator Tube Rupture (SGTR) and Main Steam Line Break (MSLB) analyses presented in Attachment 3 to this submittal.

Compliance with General Design Criterion (GDC) 19, Control Room, is demonstrated for the proposed change based on meeting the dose limit to control room personnel of 5 Rem TEDE. The original licensing basis had been established based on the whole body, thyroid, and skin dose limits now described in the Arkansas Nuclear One, Unit 1 (ANO-1) Safety Analysis Report (SAR). As stated in Regulatory Guide (RG) 1.183, the applicable acceptance criteria to establish compliance with GDC 19 for facilities licensed to use an Alternate Source Term (AST) is the 5 Rem TEDE criterion of 10 CFR 50.67(b)(2)(iii). The analysis demonstrates that ANO-1 complies with this 5 Rem TEDE requirement and, therefore, Entergy Operations, Inc. (Entergy) requests an operating license amendment to implement the new AST licensing basis.

5.2 No Significant Hazards Consideration

As provided by 10 CFR 50.67, Entergy is implementing the use of an Alternate Source Term (AST) and the dose calculation methodology described in NUREG-1465 and Regulatory Guide (RG) 1.183 to calculate accident doses to control room and offsite personnel following postulated events that result in the release of radioactive material from the reactor fuel. The AST and associated methodology define the amount, isotopic composition, physical and chemical characteristics, and timing of radioactive material releases following postulated events. Transport of the material to the control room and offsite is modeled, and the resulting Total Effective Dose Equivalent (TEDE) is determined. Regulatory acceptance criteria account for the sum of the deep-dose equivalent (for external exposures) and the committed effective dose equivalent (for internal exposures). In accordance with 10 CFR 50.67(b), licensees wishing to adopt an AST must apply for a license amendment in accordance with 10 CFR 50.90.

In support of the revised analysis applying AST, Technical Specification limits for Reactor Coolant System (RCS) and Secondary System specific activity are being reduced as part of this amendment request.

Entergy has evaluated whether a significant hazards consideration is involved with the proposed amendment by focusing on the three standards set forth in 10 CFR 50.92, "Issuance of Amendment," as discussed below:

1. Does the proposed change involve a significant increase in the probability or consequences of an accident previously evaluated?

   Response: No.

   The use of an AST is recognized in 10 CFR 50.67 and guidance for its implementation is provided in RG 1.183. The AST involves quantities, isotopic composition, chemical and physical characteristics, and release timing of radioactive material for use as inputs to accident dose analyses. As such, the AST cannot affect the probability of occurrence of a previously evaluated accident. In addition, the reduction is specific activity limits within the TSs is unrelated to accident initiators. No facility equipment, procedure, or process changes are required in conjunction with implementing the AST that could increase the likelihood of a previously analyzed accident. The proposed changes in the source term and the methodology for the dose consequence analyses follow the guidance of RG 1.183. As a result, there is no increase in the likelihood of existing event initiators.

   Regarding accident consequences, the reduction in specific activity limits within the TSs is more restrictive (more conservative) and acts to support the analysis results given the application of an AST. The results of accident dose analyses using the AST are compared to TEDE acceptance criteria that account for the sum of deep dose equivalent (for external exposure) and committed effective dose equivalent (for internal exposure). Dose results were previously compared to separate limits on whole body, thyroid, and skin doses as appropriate for the particular accident analyzed. The results of the revised dose consequences analyses demonstrate that the regulatory

acceptance criteria are met for each analyzed event. Implementing the AST involves no facility equipment, procedure, or process changes that could affect the radioactive material actually released during an event. Consequently, no conditions have been created that could significantly increase the consequences of any of the events being evaluated.

Therefore, the proposed change does not involve a significant increase in the probability or consequences of any of the events being evaluated.

2. Does the proposed change create the possibility of a new or different kind of accident from any accident previously evaluated?

Response: No.

The AST involves quantities, isotopic composition, chemical and physical characteristics, and release timing of radioactive material for use as inputs to accident dose analyses. As such, the AST cannot create the possibility of a new or different kind of accident. In addition, the reduction is specific activity limits within the TSs is unrelated to accident initiators. No facility equipment, procedure, or process changes have been made in conjunction with implementing the AST that could initiate or substantially alter the progression of an accident.

Therefore, the proposed change does not create the possibility of a new or different kind of accident from any previously evaluated.

3. Does the proposed change involve a significant reduction in a margin of safety?

Response: No.

Implementing the AST is relevant only to calculated accident dose consequences. The results of the revised dose consequences analyses demonstrate that the regulatory acceptance criteria are met for each analyzed event. In addition, the reduction is specific activity limits within the TSs is unrelated to accident initiators. No facility equipment, procedure, or process changes are required in conjunction with implementing the AST that could increase the exposure of control room or offsite individuals to radioactive material. The AST does not affect the transient behavior of non-radiological parameters (e.g., Reactor Coolant System pressure, Containment pressure) that are pertinent to a margin of safety.

Therefore, the proposed change does not involve a significant reduction in a margin of safety.

Based on the above, Entergy concludes that the proposed amendment(s) present no significant hazards consideration under the standards set forth in 10 CFR 50.92(c), and, accordingly, a finding of "no significant hazards consideration" is justified.

5.3 Environmental Considerations

The proposed amendment does not involve (i) a significant hazards consideration, (ii) a significant change in the types or significant increase in the amounts of any effluent that may be released offsite, or (iii) a significant increase in individual or cumulative occupational radiation exposure.

Concerning the types and amounts of effluents that may be released offsite, the AST involves some changes in assumed quantities and characteristics of radioactive material that are inputs to offsite accident dose calculations. These are changes to calculation assumptions only. No facility equipment, procedure, or process changes are associated with use of the AST that affect actual releases. Consequently, implementation of the AST will not increase the quantities or alter the types of radioactive material actually released if an event were to occur. Implementation of the AST also has no effect on the actual or calculated effluents arising from normal operation.

With respect to occupational doses, the AST only involves a change in accident dose calculation inputs and methodology. Calculated doses meet TEDE criteria. No aspect of implementing the AST involves facility equipment, procedure, or process changes that would increase actual onsite doses if an event were to occur. The AST does not result in actual or calculated changes in the normal radiation levels in the facility, or in the type or quantity of radioactive materials processed during normal operation.

Accordingly, the proposed amendment meets the eligibility criteria for categorical exclusion set forth in 10 CFR 51.22(c)(9). Therefore, pursuant to 10 CFR 51.22(b), no environmental impact statement or environmental assessment need be prepared in connection with this proposed amendment.

6.0 PRECEDENCE

Several plants have been approved for the use of the AST, including Grand Gulf Nuclear Station in March 2001, D. C. Cook in November 2002, and River Bend Station in March 2003. Those submittals included various TS changes which were justified by the change to the AST. This ANO-1 license amendment request also involves changes to the TSs as identified in Section 2.0.

7.0 REFERENCES

1. NRC letter dated June 12, 2003, *Generic Letter 2003-01, Control Room Habitability* (0CNA060308)

2. NUREG-1465, *Accident Source Terms for Light Water Nuclear Power Plants*, L. Soffer, et. al., February 1995

3. Regulatory Guide 1.183, *Alternative Radiological Source Terms for Evaluating Design Basis Accidents at Nuclear Power Reactors*, USNRC, July 2000

4. Federal Register Notice dated January 17, 2007, *Notice of Availability of Technical Specification Improvement To Modify Requirements Regarding Control Room Envelope Habitability Using the Consolidated Line Item Improvement Process* (*FRN* Federal Register, Vol. 72, No. 10, page 2022)

5. Entergy Letter dated August 28, 2003, *Response to Generic Letter 2003-01* (0CAN080304)

6. Entergy Letter dated June 17, 2004, *Proposed Operating License Amendment for Revised ANO-1 Control Room Habitability Analysis* (1CAN060401)

7. Entergy letter dated May 6, 2006, *Withdrawal of Proposed Operating License Amendment for Revised ANO-1 Control Room Habitability Analysis* (1CAN050601)

8. Entergy Letter dated February 14, 2007, *Supplemental Response to Generic Letter 2003-01 Regarding Control Room Habitability* (0CAN020701)

Attachment 2

1CAN100703

Proposed Technical Specification Changes (mark-up)

3.4 REACTOR COOLANT SYSTEM (RCS)

3.4.12 RCS Specific Activity

LCO 3.4.12 The specific activity of the reactor coolant shall be:

a. ≤ 1.0~~3.5~~ µCi/gm DOSE EQUIVALENT I-131; and

b. ≤ 72/$\bar{E}$ µCi/gm total.

APPLICABILITY: MODES 1 and 2,
MODE 3 with RCS average temperature (T_{avg}) ≥ 500°F.

ACTIONS

CONDITION	REQUIRED ACTION	COMPLETION TIME
A. Specific activity not within limits.	A.1 Restore specific activity to within limit(s).	24 hours
B. Required Action and associated Completion Time not met. OR DOSE EQUIVALENT I-131 > 60 µCi/gm.	B.1 Be in MODE 3 with T_{avg} < 500°F.	6 hours

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.4.12.1	Verify reactor coolant gross specific activity ≤ 72/$\bar{E}$ µCi/gm.	7 days

SURVEILLANCE		FREQUENCY
SR 3.4.12.2	----------NOTE---------- Only required to be performed in MODE 1. ---------- Verify reactor coolant DOSE EQUIVALENT I-131 specific activity ≤ 1.0~~3.5~~ µCi/gm.	14 days
SR 3.4.12.3	----------NOTE---------- Not required to be performed until 31 days after a minimum of 2 EFPD and 20 days of MODE 1 operation have elapsed since the reactor was last subcritical for ≥ 48 hours. ---------- Determine $\bar{E}$.	184 days

3.7 PLANT SYSTEMS

3.7.4 Secondary Specific Activity

LCO 3.7.4 The specific activity of the secondary coolant shall be ≤ 0.1~~7~~ μCi/gm DOSE EQUIVALENT I-131.

APPLICABILITY: MODES 1, 2, 3, and 4.

ACTIONS

CONDITION	REQUIRED ACTION	COMPLETION TIME
A. Specific activity not within limit.	A.1 Be in MODE 3.	6 hours
	AND	
	A.2 Be in MODE 5.	36 hours

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.7.4.1	Verify the specific activity of the secondary coolant is ≤ 0.1~~7~~ μCi/gm DOSE EQUIVALENT I-131.	31 days

Attachment 3

1CAN100703

Summary Tables of Dose Analyses Input

Table 1

ANO-1 Source Terms for LOCA

The following table presents the core radionuclide inventory at current rated power, including 2% uncertainty (2619.36 MWt). This isotopic inventory has been calculated using the ORIGEN-S code. These values were utilized in the LOCA dose analyses.

Isotope	Core Inventory [Curies]	Isotope	Core Inventory [Curies]	Isotope	Core Inventory [Curies]
Kr-85	9.61E+05	**Sb-127**	6.56E+06	**Ce-143**	1.12E+08
Kr-85m	1.90E+07	**Sb-129**	2.01E+07	**Ce-144**	1.05E+08
Kr-87	3.73E+07	**Te-127**	6.52E+06	**Np-239**	1.39E+09
Kr-88	5.01E+07	**Te-127m**	1.16E+06	**Pu-238**	1.93E+05
Xe-131m	7.55E+05	**Te-129**	1.88E+07	**Pu-239**	2.51E+04
Xe-133	1.48E+08	**Te-129m**	3.66E+06	**Pu-240**	3.88E+04
Xe-133m	4.60E+06	**Te-131m**	1.40E+07	**Pu-241**	9.82E+06
Xe-135	3.51E+07	**Te-132**	1.02E+08	**Am-241**	1.02E+04
Xe-135m	3.09E+07	**Sr-89**	7.25E+07	**Cm-242**	2.71E+06
Xe-138	1.27E+08	**Sr-90**	7.47E+06	**Cm-244**	1.99E+05
I-130	1.36E+06	**Sr-91**	8.78E+07	**La-140**	1.32E+08
I-131	7.22E+07	**Sr-92**	9.40E+07	**La-142**	1.15E+08
I-132	1.05E+08	**Ba-139**	1.32E+08	**Nb-95**	1.34E+08
I-133	1.48E+08	**Ba-140**	1.28E+08	**Nd-147**	4.70E+07
I-134	1.67E+08	**Mo-99**	1.35E+08	**Pr-143**	1.11E+08
I-135	1.41E+08	**Rh-105**	7.25E+07	**Y-90**	7.75E+06
Cs-134	1.46E+07	**Ru-103**	1.14E+08	**Y-91**	9.53E+07
Cs-136	2.98E+06	**Ru-105**	7.64E+07	**Y-92**	9.51E+07
Cs-137	9.88E+06	**Ru-106**	4.19E+07	**Y-93**	1.07E+08
Cs-138	1.38E+08	**Tc-99m**	1.18E+08	**Zr-95**	1.29E+08
Rb-86	1.29E+05	**Ce-141**	1.23E+08	**Zr-97**	1.23E+08

Table 2

ANO-1 Source Terms for CREA

The following table presents the core radionuclide inventory at current rated power, including 2% uncertainty (2619.36 MWt). This isotopic inventory has been calculated using the ORIGEN-S code. These values were utilized in the CREA dose analyses.

Isotope	Core Inventory [Curies]
Kr-83m	8.77E+06
Kr-85	9.61E+05
Kr-85m	1.90E+07
Kr-87	3.73E+07
Kr-88	5.01E+07
Xe-131m	7.55E+05
Xe-133	1.48E+08
Xe-133m	4.60E+06
Xe-135	3.51E+07
Xe-135m	3.09E+07
Xe-138	1.27E+08
I-130	1.36E+06
I-131	7.22E+07
I-132	1.05E+08
I-133	1.48E+08
I-134	1.67E+08
I-135	1.41E+08
Cs-134	1.46E+07
Cs-136	2.98E+06
Cs-137	9.88E+06
Cs-138	1.38E+08
Rb-86	1.29E+05

Table 3

ANO-1 Source Terms for FHA

The following table presents the core radionuclide inventory at current rated power, including 2% uncertainty (2619.36 MWt). This isotopic inventory has been calculated using the ORIGEN-S code. These values were utilized in the FHA dose analyses.

Isotope	Core Inventory [Curies]
Kr-85	9.61E+05
Kr-85m	1.90E+07
Kr-87	3.73E+07
Kr-88	5.01E+07
Xe-131m	7.55E+05
Xe-133	1.48E+08
Xe-133m	4.60E+06
Xe-135	3.51E+07
Xe-135m	3.09E+07
Xe-138	1.27E+08
I-130	1.36E+06
I-131	7.22E+07
I-132	1.05E+08
I-133	1.48E+08
I-134	1.67E+08
I-135	1.41E+08
Sb-131	5.67E+07
Te-131	6.10E+07
Te-131m	1.40E+07
Te-132	1.02E+08
Te-133	7.89E+07
Te-133m	7.02E+07

Table 4

ANO-1 Source Terms for SGTR & MSLB

This table presents the RCS and secondary radionuclide inventories for use in the SGTR and MSLB dose analyses. The RCS activity is based on an equilibrium 1 µCi/g dose equivalent I-131 and 72/E µCi/g total. The secondary activity is based on an equilibrium 0.1 µCi/g dose equivalent I-131.

Isotope	RCS Activity (Ci)	Secondary Activity (Ci)
Kr85	4.80E+02	0.00E+00
Kr85m	1.31E+03	0.00E+00
Kr87	2.09E+03	0.00E+00
Kr88	2.93E+03	0.00E+00
Xe131m	3.78E+02	0.00E+00
Xe133	3.02E+04	0.00E+00
Xe133m	7.64E+02	0.00E+00
Xe135	1.30E+04	0.00E+00
Xe135m	1.18E+03	0.00E+00
Xe-138	3.46E+03	0.00E+00
I130	6.82E+02	1.29E+01
I131	7.33E+01	1.39E+00
I132	1.00E+03	1.90E+01
I133	6.91E+02	1.31E+01
I134	1.48E+03	2.81E+01
I135	1.22E+03	2.31E+01
Cs-134	5.11E+02	9.70E+00
Cs-136	4.03E+01	7.66E-01
Cs-137	4.22E+02	8.01E+00
Cs-138	1.00E+04	1.90E+02
Rb-86	6.43E+01	1.22E+00

Table 5

Generic Input Parameters

Parameter	Input Value
Power level for analysis (102%)	2619.36 MWt
RB Net Free Volume	1.81E6 ft^3
RB Leak Rates	0.2%/day ≤ 24 hrs 0.1%/day > 24 hrs
Offsite χ/Q	6.8E-4 s/m^3 0-720 hrs (EAB) 1.1E-4 s/m^3 0-8 hrs (LPZ) 1.1E-5 s/m^3 8-24 hrs (LPZ) 4.0E-6 s/m^3 24-96 hrs (LPZ) 1.3E-6 s/m^3 > 96 hrs (LPZ)
Offsite Breathing Rates	3.5E-4 m^3/s 0-8 hrs 1.8E-4 m^3/s 8-24 hrs 2.3E-4 m^3/s > 24 hrs
CR Breathing Rate	3.5E-4 m^3/s for duration of the event
CR Occupancy Factors	1.0 0-24 hrs 0.6 1-4 days 0.4 > 4 days
CR Volume	40,000 ft^3
CR Filtered Inleakage	333 cfm
CR Recirculation Flow	1667 cfm
CR Intake Filter Efficiency (two 2" filters)	99% for particulate (aerosols) and for elemental and organic iodine
CR Recirculation Filter Efficiency (one 2" filter)	99% for particulate (aerosols) and 95% for elemental and organic iodine
Dose Conversion Factors (DCF)	Federal Guidance Report 11 CEDE and Federal Guidance Report 12 EDE

Table 6

LOCA Input Parameters

Parameter	Input Value
Gap Release Phase	30 sec – 0.5 hrs
Early In-Vessel Release Phase	0.5 – 1.8 hrs
Gap Release Fraction	0.05 for noble gases, halogens, and alkali metals only
Early In-Vessel Release Fractions	0.95 noble gases 0.35 halogens 0.25 alkali metals 0.05 tellurium metals 0.02 strontium and barium 0.0025 noble metals 0.0005 cerium group 0.0002 lanthanides
Iodine species distribution (%)	95.00 particulate 4.85 elemental 0.15 organic
Unsprayed Volume	2.00E5 ft^3 (rounded up)
Sump Volume	54,918 ft^3
Sprayed Volume	1.61E6 ft^3
Sprayed Fractions	0.11 unsprayed 0.89 sprayed
RB Mixing Rates	6270 cfm unsprayed to sprayed 6270 cfm sprayed to unsprayed
Amount of RB Leakage into Penetration Rooms	50%
Penetration Room Ventilation System Filter Efficiency	99% particulates 90% elemental and organic iodines 0% noble gases
Spray Removal Rates Elemental organic particulate	 20 hr^{-1} during injection, 10 hr^{-1} during recirculation until DF=200, then 0 No removal 2.60 hr^{-1} until DF=50, then 0.26 hr^{-1} until DF=1000, then 0
Spray Initiation	300 sec
CR χ/Q	4.46E-3 s/m^3 0-2 hrs 3.05E-3 s/m^3 2-8 hrs 1.36E-3 s/m^3 8-24 hrs 8.70E-4 s/m^3 24-96 hrs 7.36E-4 s/m^3 >96 hrs
CR Unfiltered Inleakage	82 cfm

Table 6 (continued)

ESF Leakage Input for LOCA Analyses

Parameter	Input Value
ESF Leakage Rate	782 cc/hr (4.603E-4 cfm)
Fraction of Released Iodine in Sump Solution	1.0
Iodine Species Distribution in Sump	0.97 elemental 0.03 organic
Time to Recirculation	4257 sec (1.1825 hr)
Iodine Partition Coefficient of ESF Leakage	0.10

Table 7

CREA Input Parameters

Parameter	Input Value
Fuel Failure (rods in DNB)	14%
Peaking Factor	1.8
Fission Product Gap Fractions	0.10 noble gases and iodines 0.12 alkali metals
RB Release Iodine Species Distribution	95% particulate 4.85% elemental 0.15% organic
Secondary Release Iodine Species Distribution	0% particulate 97% elemental 3% organic
Primary-to-Secondary (P-S) Leak Rate (secondary release model)	300 gpd
Duration of Secondary Release Event (switch to DHR system)	38.25 hrs
Flashing and Vaporizing Fraction of P-S Leakage during Cooldown (no partitioning)	0.15
Sedimentation Coefficient	0.1/hr until DF = 1000, then 0
Partition Coefficients of P-S Leakage Mixed with Secondary Liquid Inventory	0.01 iodines 0.001 alkali metals
Steam Release Rates from Secondary	2.5815E+6 g/min 0-2 hrs 5.6977E+5 g/min 2-38.25 hrs
CR χ/Q (RB release)	3.55E-3 s/m^3 0-2 hrs 2.49E-3 s/m^3 2-8 hrs 9.85E-4 s/m^3 8-24 hrs 8.30E-4 s/m^3 24-96 hrs 6.31E-4 s/m^3 > 96 hrs
CR χ/Q (secondary release)	1.90E-2 s/m^3 0-0.5 hrs 4.10E-3 s/m^3 0.5-2 hrs 2.59E-3 s/m^3 2-8 hrs 1.12E-3 s/m^3 8-24 hrs 8.32E-4 s/m^3 24-96 hrs 5.91E-4 s/m^3 > 96 hrs
CR Unfiltered Inleakage	82 cfm

Table 8

SGTR Input Parameters

Parameter	Input Value
Initial Primary Coolant Activity	1 µCi/g I-131
Activity with Pre-existing Iodine Spike	60 µCi/g I-131
Initial Secondary Coolant Activity	0.1 µCi/g I-131
Accident-Initiated Iodine Spike Factor	335
Accident-Initiated Iodine Spike Duration	8 hrs
Initial Ruptured SG Tube Leak Rate	435 gpm
Primary-to-Secondary Leak Rate	150 gpd per steam generator
Time to Reactor Trip (full steaming until trip)	11 min
Time to Isolation of Faulted SG	34 min
Time to Isolation of Intact SG (initiation of DHR)	237.8 hrs
Flashing Fraction in Faulted SG	0.15
Partition Coefficients prior to Reactor Trip (release via condenser)	0.0001 iodines and alkali metals
Partition Coefficients after Reactor Trip (release via MSSV or ADV)	0.01 iodines 0.001 alkali metals
CR χ/Q	4.10E-3 s/m^3 0-11 min (reactor trip) 1.90E-2 s/m^3 11-34 min (faulted SG isolated) 4.10E-3 s/m^3 0.5667-2 hrs 2.59E-3 s/m^3 2-8 hrs 1.12E-3 s/m^3 8-24 hrs 8.32E-4 s/m^3 24-96 hrs 5.91E-4 s/m^3 > 96 hrs
CR Unfiltered Inleakage	85 cfm

Table 9

MSLB Input Parameters

Parameter	Input Value
Initial Primary Coolant Activity	1 µCi/g I-131
Activity with Pre-existing Iodine Spike	60 µCi/g I-131
Initial Secondary Coolant Activity	0.1 µCi/g I-131
Accident-Initiated Iodine Spike Factor	500
Accident-Initiated Iodine Spike Duration	8 hrs
Primary-to-Secondary Leak Rate	0.5 gpm/SG
Time to Begin Cooldown (operator action)	30 min
Time to Isolation of Unaffected SG (initiation of DHR)	237.8 hrs
Time to Reach 212 F	251.8 hrs
Faulted SG Mass	6.00E+4 lbm
Flashing Fraction in Unaffected SG	0.2
Partition Coefficient (faulted SG and intact SG via flashing)	1.0
Partition Coefficients (intact SG via steaming)	0.01 iodines 0.001 alkali metals
CR χ/Q (faulted SG – main steam line)	3.15E-3 s/m^3 0-2 hrs 2.16E-3 s/m^3 2-8 hrs 8.90E-4 s/m^3 8-24 hrs 6.61E-4 s/m^3 24-96 hrs 5.01E-4 s/m^3 > 96 hrs
CR χ/Q (intact SG – MSSV/ADV)	1.90E-2 s/m^3 0-0.5 hrs 4.10E-3 s/m^3 0.5-2 hrs 2.59E-3 s/m^3 2-8 hrs 1.12E-3 s/m^3 8-24 hrs 8.32E-4 s/m^3 24-96 hrs 5.91E-4 s/m^3 > 96 hrs
CR Unfiltered Inleakage	85 cfm

Table 10

FHA Input Parameters

Parameter	Input Value
Peaking Factor	1.8
Offsite and CR Breathing Rate (duration of event)	3.5×10^{-4} m^3/s
Offsite χ/Q (duration of event)	6.8×10^{-4} s/m^3 EAB 1.1×10^{-4} s/m^3 LPZ
Control Room χ/Q (containment more limiting than fuel handling area ventilation) (duration of event)	3.55×10^{-3} s/m^3
Gap Fractions Released Kr-85 I-131 (modified per NUREG/CR-5009) Other isotopes	 0.30 0.12 0.10
Time after Shutdown	72 hrs
Number of Damaged Rods	82 (six rows)
Fuel Rod Pressure Limit	1500 psig
Iodine Form Elemental Organic	 99.85% 0.15%
Pool Decontamination Factors Elemental Iodine Organic Iodine and Noble Gases	 286 (limited to provide overall DF = 200) 1
CR Unfiltered Inleakage	85 cfm

Attachment 4

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Summary Results of Dose Analyses

LOCA TEDE Doses (rem)

	EAB (Worst 2 hr Dose)	LPZ (0-30 days)	CR (82 cfm) (0-30 days)
TEDE Dose (rem)	**10.49**	**2.56**	**3.77**
Acceptance Criteria (rem)	25.0	25.0	5.0

CREA Containment Leakage TEDE Doses (rem)

	EAB (Worst 2 hr Dose)	LPZ (0-30 days)	CR (82 cfm) (0-30 days)
TEDE Dose (rem)	**4.73**	**2.28**	**3.40**
Acceptance Criteria (rem)	6.3	6.3	5.0

CREA Primary to Secondary Leakage TEDE Doses (rem)

	EAB (Worst 2 hr Dose)	LPZ (Event Duration)	CR (82 cfm) (0-30 days)
TEDE Dose (rem)	**3.03**	**1.64**	**4.95**
Acceptance Criteria (rem)	6.3	6.3	5.0

SGTR Accident Initiated Iodine Spike TEDE Doses (rem)

	EAB (Worst 2 hr Dose)	LPZ (Event Duration)	CR (85 cfm) (0-30 days)
TEDE Dose (rem)	**1.26**	**0.23**	**1.00**
Acceptance Criteria (rem)	2.5	2.5	5.0

SGTR Pre-Existing Iodine Spike TEDE Doses (rem)

	EAB (Worst 2 hr Dose)	LPZ (Event Duration)	CR (85 cfm) (0-30 days)
TEDE Dose (rem)	**2.20**	**0.37**	**2.33**
Acceptance Criteria (rem)	25.0	25.0	5.0

MSLB Accident Initiated Iodine Spike TEDE Doses (rem)

	EAB (Worst 2 hr Dose)	LPZ (Event Duration)	CR (85 cfm) (0-30 days)
TEDE Dose (rem)	**2.07**	**1.05**	**3.72**
Acceptance Criteria (rem)	2.5	2.5	5.0

MSLB Pre-Existing Iodine Spike TEDE Doses (rem)

	EAB (Worst 2 hr Dose)	LPZ (Event Duration)	CR (85 cfm) (0-30 days)
TEDE Dose (rem)	**0.45**	**0.19**	**1.84**
Acceptance Criteria (rem)	25.0	25.0	5.0

FHA TEDE Doses (rem) 72 hr Decay

	EAB (Worst 2 hr Dose)	LPZ (2 hr Release)	CR (85 cfm) (0-30 days)
TEDE Dose (rem) 82 rods Damaged	**1.40**	**0.25**	**1.00**
Acceptance Criteria (rem)	6.3	6.3	5.0

Locked Reactor Coolant Pump Rotor (LR)

The LR DBA is bounded by the dose consequences calculated for the Main Steam Line Break event, since no fuel damage will occur following a LR event.